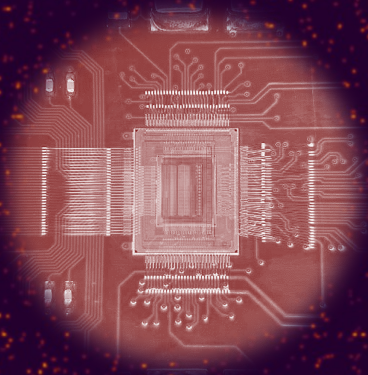
2026/10/20

Silicon bring-up considerations



Llorenç Fanals-i-Batllori[1,†]

[1]SLAC National Accelerator Laboratory

[†]fanals@slac.stanford.edu

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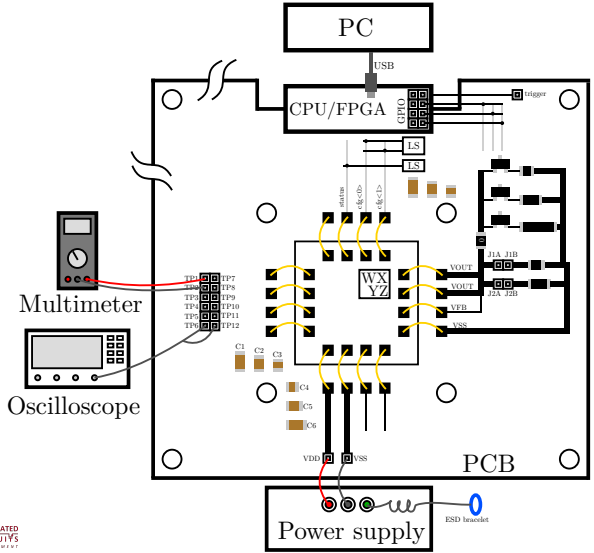
Silicon bring-up considerations

Silicon bring-up

You just received your ASICs. Now what?

1. PCB design.
2. Packaging (wire bonding).
3. Embedded system.
4. Sanity checks.
5. Power-up.
6. Characterization.

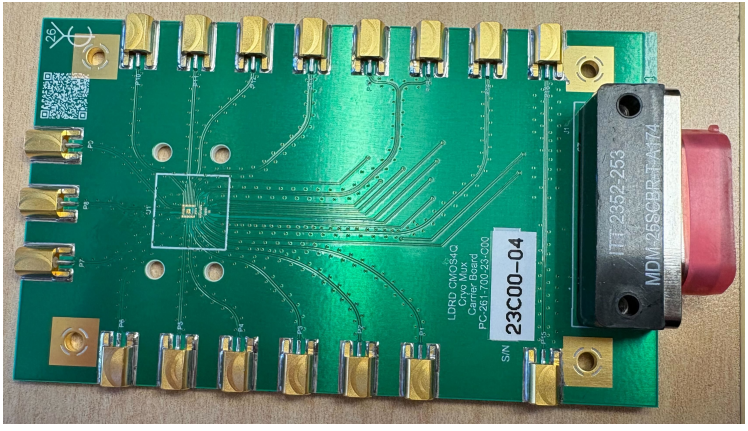
System


PC
USB
CPU/FPGA
GPIO
trigger
LS
LS
status
dg<1>
dg<0>
J1A J1B
J2A J2B
VOUT
VOUT
VFB
VSS
WX
YZ
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
C1
C2
C3
C4
C5
C6
VDD
VSS
PCB
Multimeter
Oscilloscope
Power supply
ESD bracelet

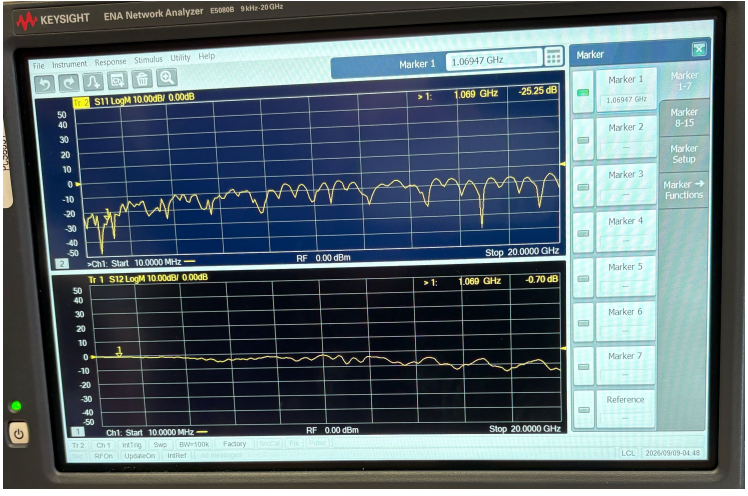
General advice

- Document.
- Use version control (git).
- Use redundancy, avoid a single point of failure.
- Print in paper your pinout (ASIC and PCB).
- Wear an ESD bracelet.
- Use AI wisely.

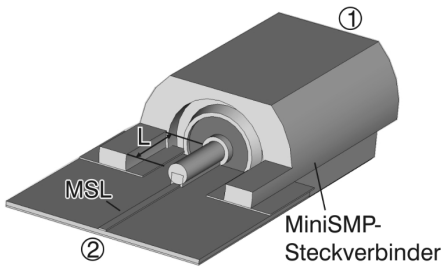
RF PCB: $5500, ROGERS 4350B



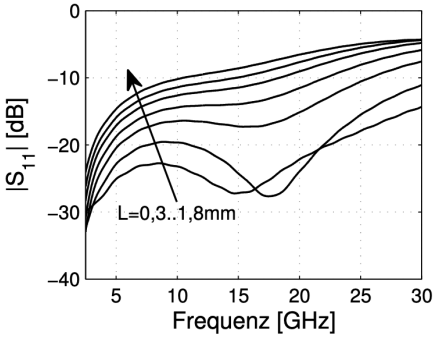
RF PCB


KEYSIGHT
ENA Network Analyzer
E5080B
9 kHz-20 GHz
File Instrument Response Stimulus Utility Help
Marker 1
1.06947 GHz
Marker
S11 LogM 10.00dB/ 0.00dB
> 1: 1.069 GHz -25.25 dB
>Ch1: Start 10.0000 MHz
RF 0.00 dBm
Stop 20.0000 GHz
Tr 1 S12 LogM 10.00dB/ 0.00dB
> 1: 1.069 GHz -0.70 dB
Ch1: Start 10.0000 MHz
RF 0.00 dBm
Stop 20.0000 GHz
Marker 1
Marker 2
Marker 3
Marker 4
Marker 5
Marker 6
Marker 7
Reference
Marker 1-7
Marker 8-15
Marker Setup
Marker → Functions
2026/09/09-04:48

Literature (TU Hamburg) [1]



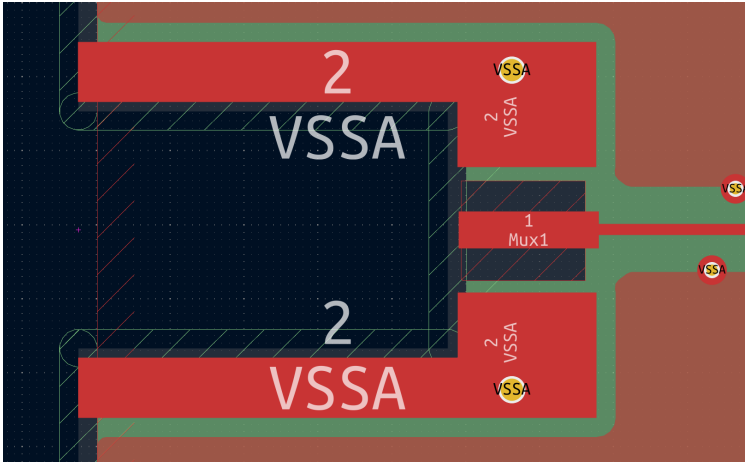
a)



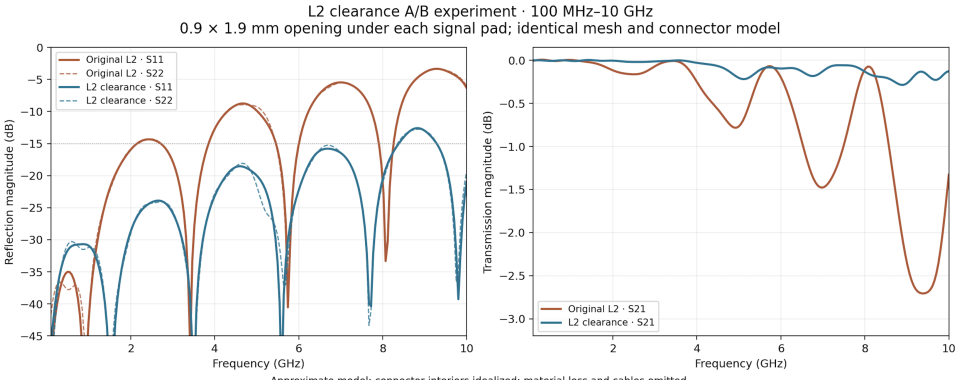
b)

Abbildung 3.36: a) Simulationsmodell des MiniSMP-MSL-Übergangs und b) simulierte Rückflussdämpfung an Tor 1.

Connector footprint

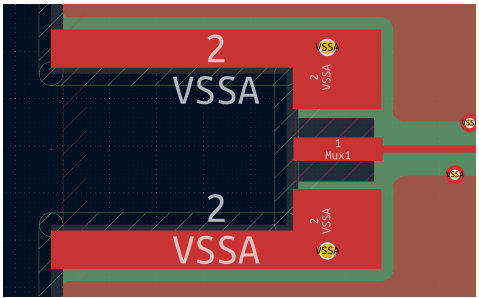


EM simulations

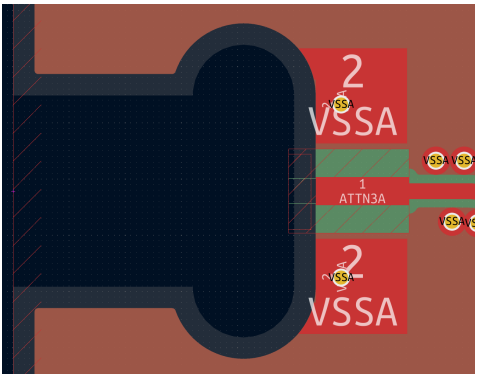


Connector footprint

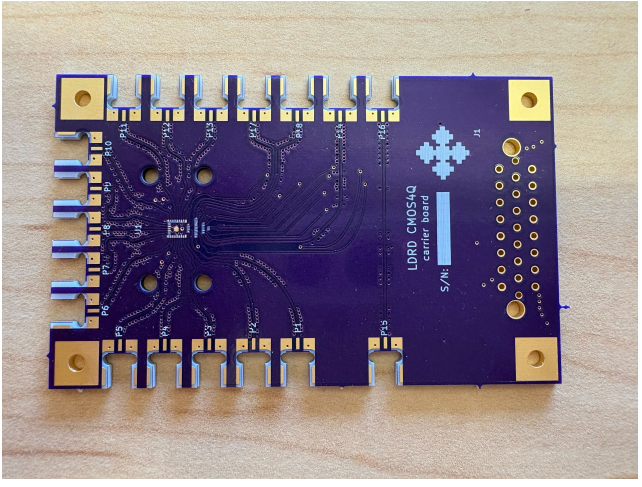
v1



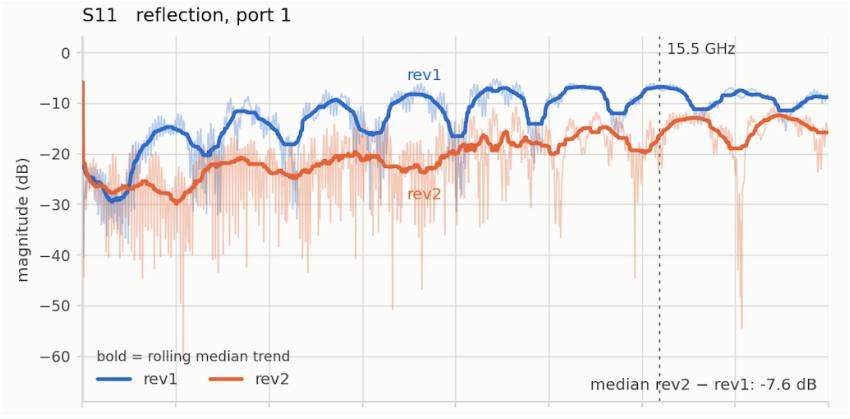
v2



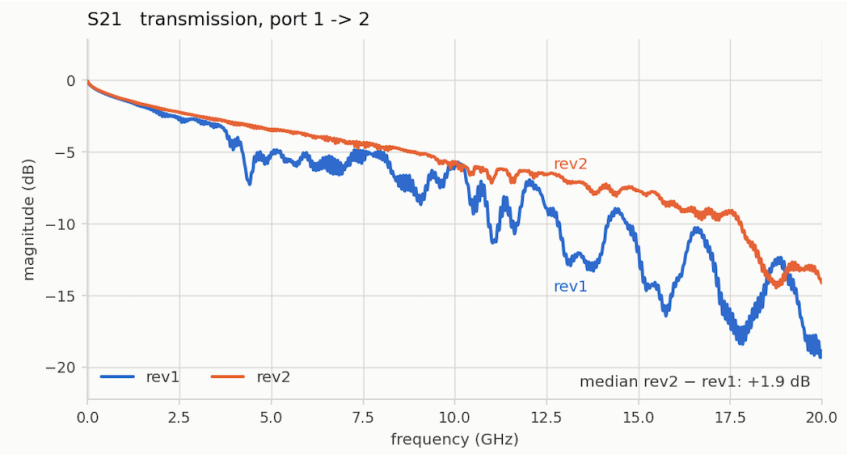
RF PCB v2: $241, FR4



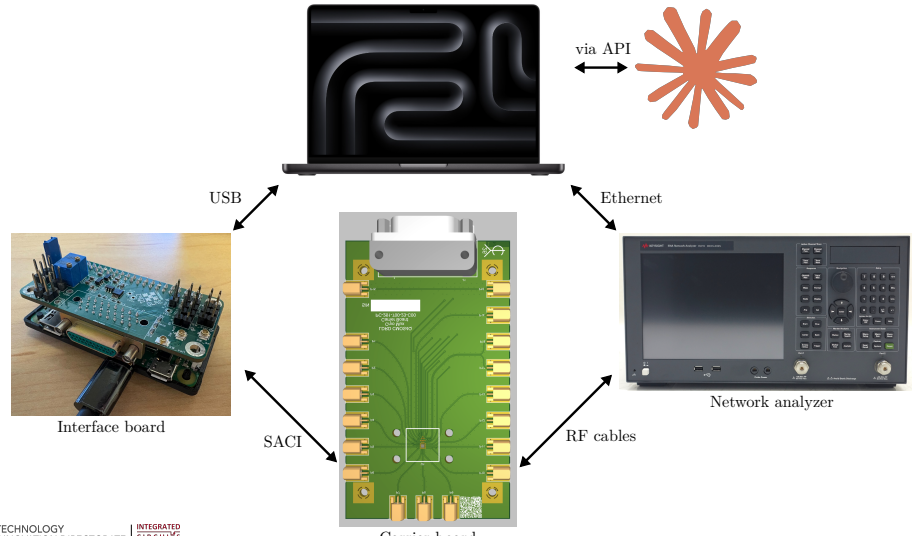
S_{11} measured



S_{21} measured



Setup


via API
USB
Ethernet
Interface board
SACI
Carrier board
RF cables
Network analyzer

Video

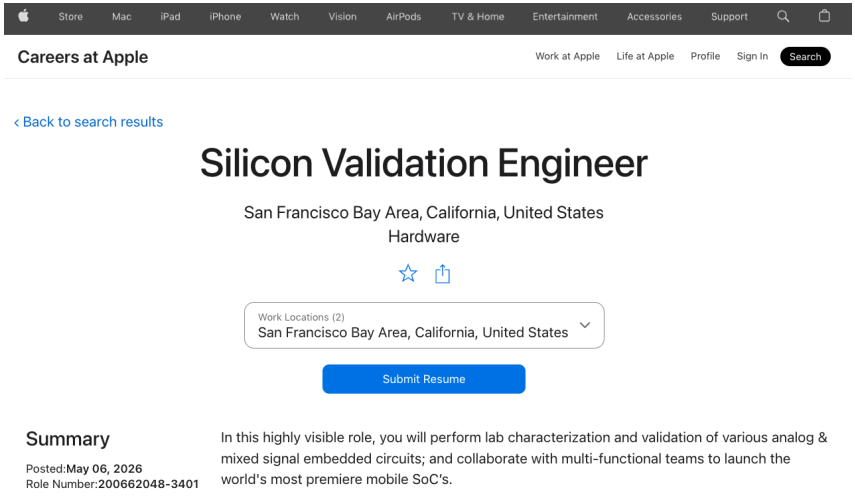
2

Backup

Silicon bring-up

You just received your ASICs. Now what?

Hand it off?


Store Mac iPad iPhone Watch Vision AirPods TV & Home Entertainment Accessories Support
Careers at Apple
Work at Apple Life at Apple Profile Sign In Search
‹ Back to search results
Silicon Validation Engineer
San Francisco Bay Area, California, United States
Hardware
Work Locations (2)
San Francisco Bay Area, California, United States
Submit Resume
Summary
Posted:May 06, 2026
Role Number:200662048-3401
In this highly visible role, you will perform lab characterization and validation of various analog & mixed signal embedded circuits; and collaborate with multi-functional teams to launch the world's most premiere mobile SoC's.

Silicon bring-up

You just received your ASICs. Now what?

1. PCB design.
2. Packaging (wire bonding).
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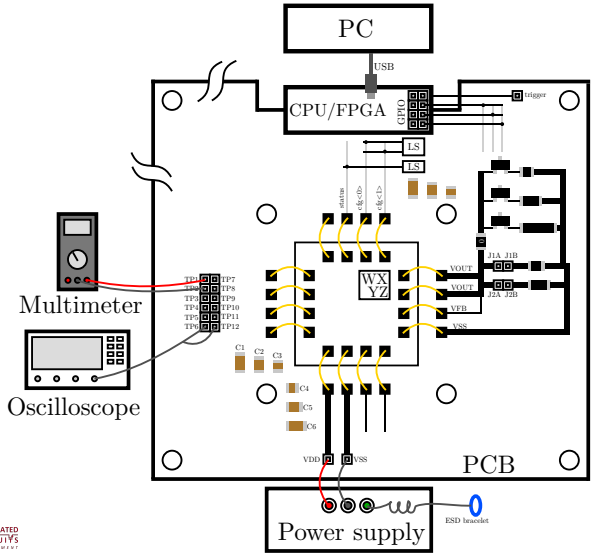
Do it yourself!



General advice

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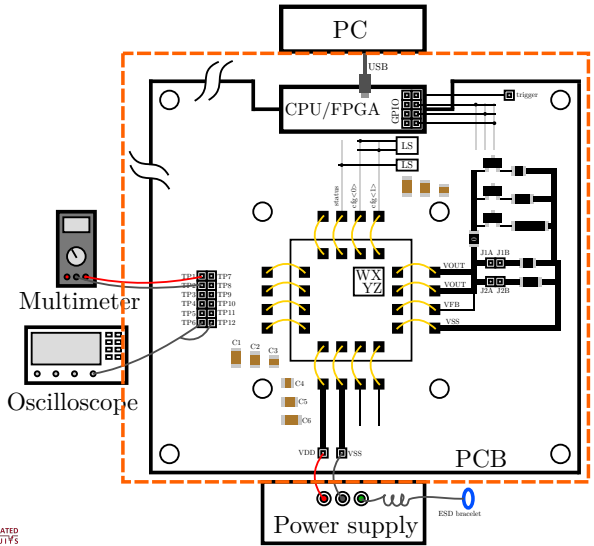
- Use version control (git).
- Use redundancy, avoid a single point of failure.
- Print in paper your pinout (ASIC and PCB).
- Wear an ESD bracelet.

3

PCB design

PCB design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USB
CPU/FPGA
GPIO
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LS
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status
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J2A J2B
VOUT
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VFB
VSS
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TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
C1
C2
C3
C4
C5
C6
VDD
VSS
PCB
Multimeter
Oscilloscope
Power supply
ESD bracelet

PCB design

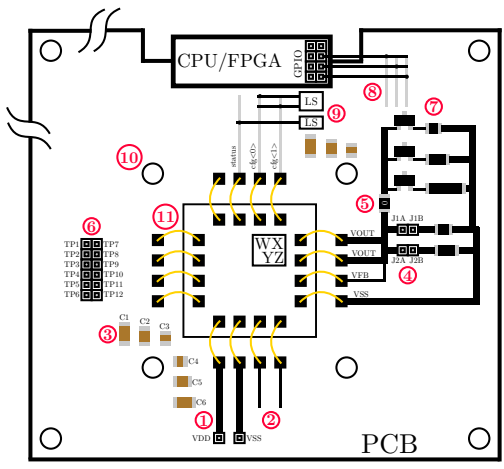
Potentials software to use:

- Altium: profressional-grade, needs a license.
- KiCad: ”hobbyist” software but extremely capable as of 2026.
- EasyEDA: limited capabilities but smooth integration with PCB manufacturer.

PCB design I

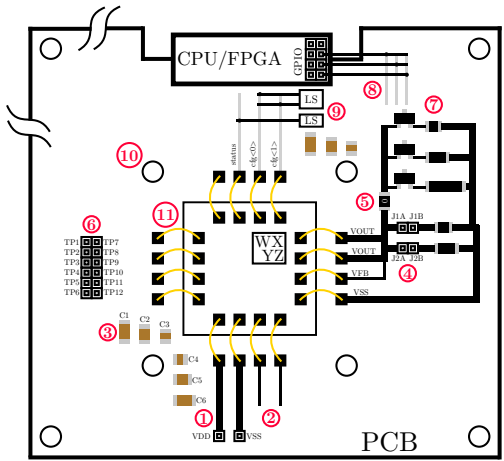
Obtain the sheet resistance of the conducting metal, in $\frac{\Omega}{\square}$.

1. Design to minimize voltage drop *IR* on nets that carry current.
2. Nets driving a high-impedance pin (transistor gate) are much more relaxed. Make them minimum dimensions ($\sim$ 4 mil).



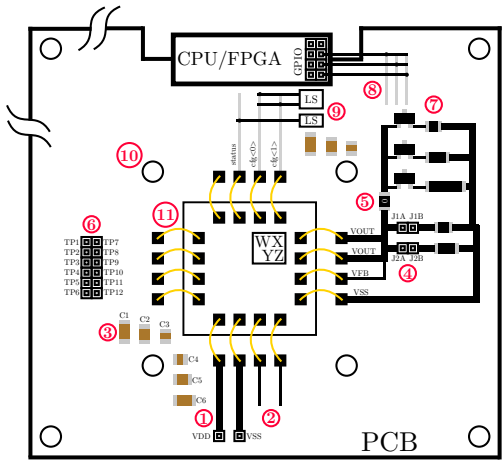
PCB design II

3. Use decoupling capacitors between supply and ground. Use different cap values to improve the frequency response (or not [2]).
4. Use jumpers to increase flexibility during testing.
5. Or 0-Ohm resistors.
6. Place extensive test points. There must be a good reason for a PCB net to not be exposed/measurable. Label them.



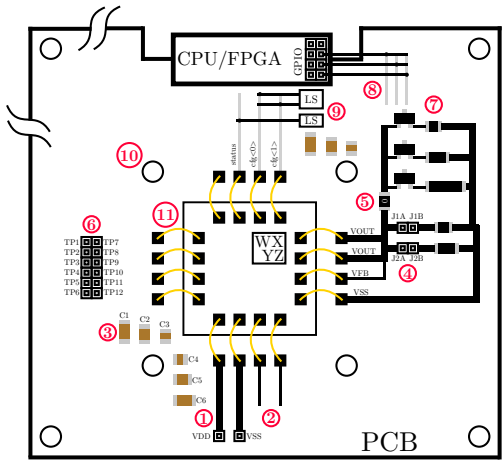
PCB design III

7. You can place an array of binary-weighted resistors and small R_{DS} MOSFETs to realize a tunable load. The gate voltages may need a level shifter (LS). Be aware of the GPIO latency.
8. Use metals at different levels in an orthogonal fashion.
9. Level shifters (LS) are needed to accommodate the ASIC voltage domain to that of the GPIOs, and viceversa.

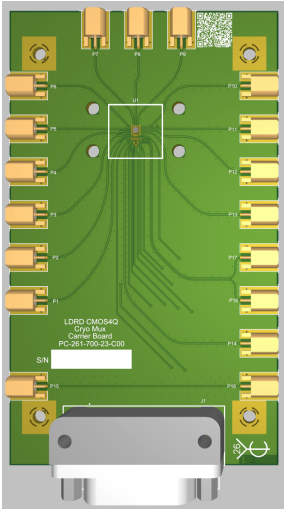


PCB design IV

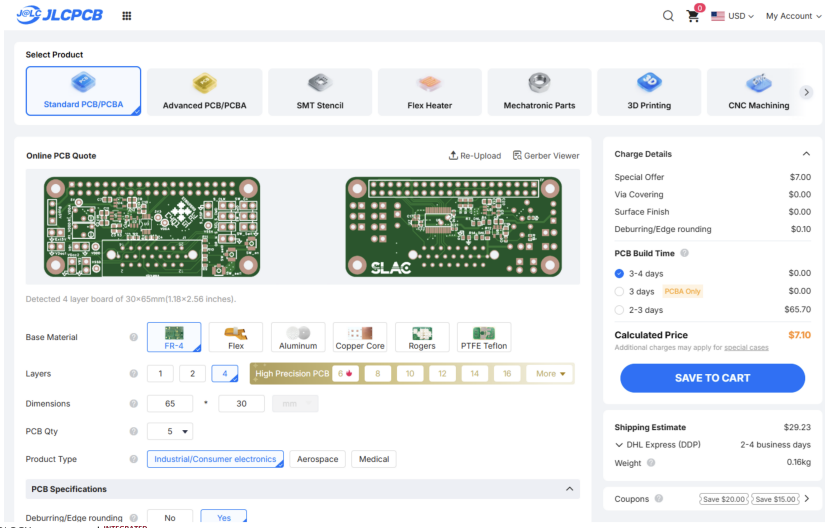
10. Place mounting holes both at the PCB level and around the ASIC.
11. Give a separation between PCB pads and ASIC of 300 μm at least. A special surface finish may be needed.



PCB design: example I (Andrew Young)

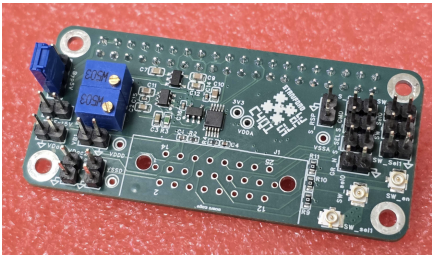


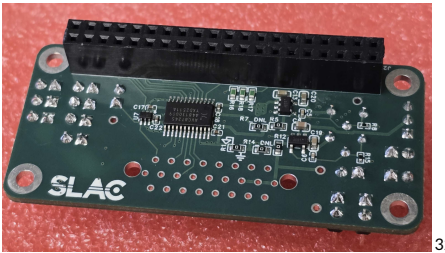
PCB design: example II (Joan Serrano Aporta)



PCB design: example II (Joan Serrano Aporta)

- Total cost: $\sim$ 350$, 5 boards, entirely populated.
- Turnaround time: 1 week.
- No issues observed.

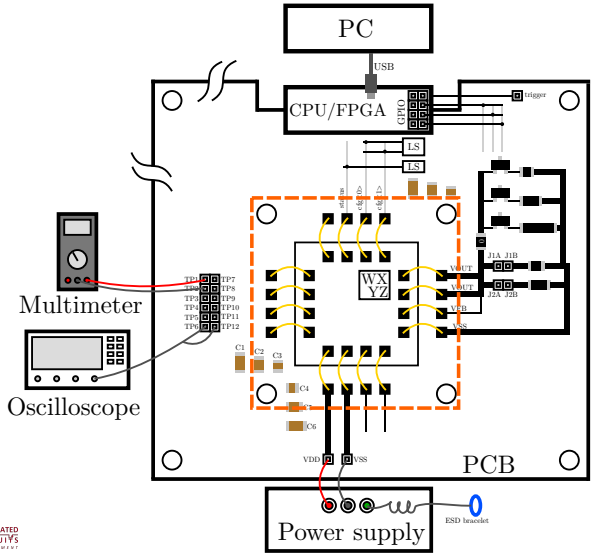




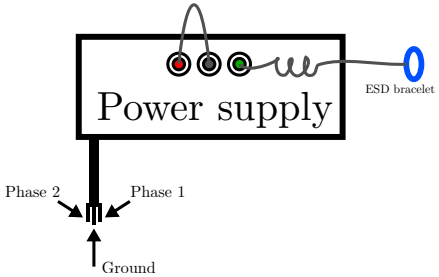
4

Packaging (wire bonding)

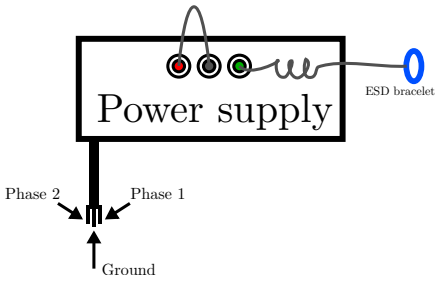
Packaging (wire bonding)

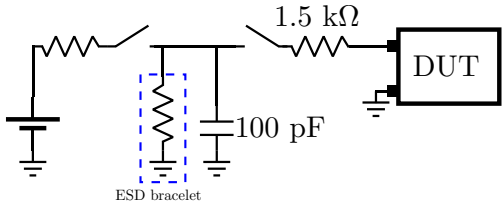


ESD bracelet



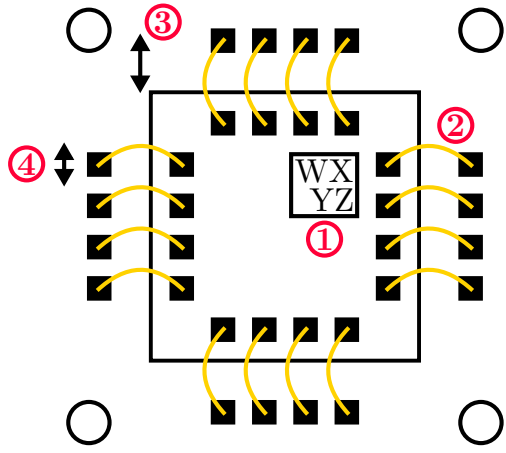
ESD bracelet



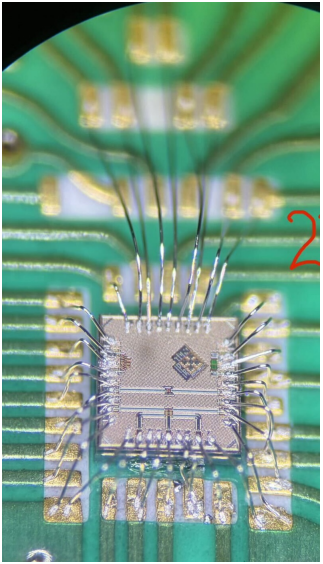


Packaging (wire bonding)

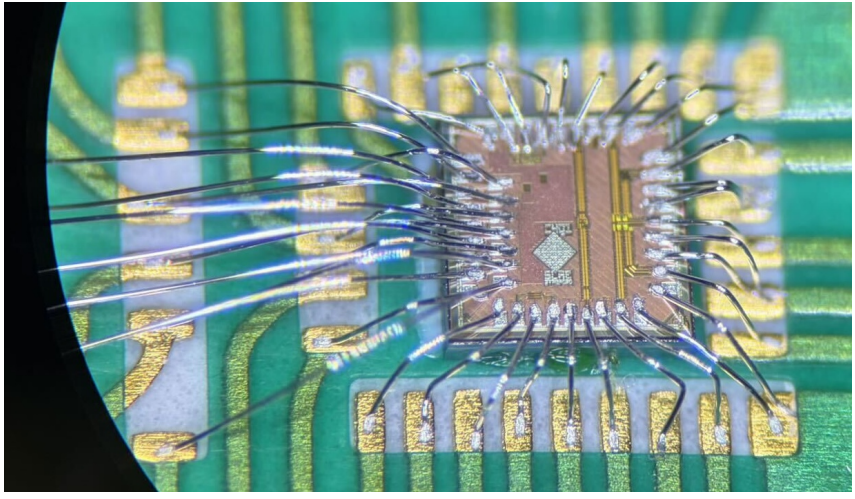
1. Give clear indications of the ASIC orientation. A logo or a disctinctive feature can be extremely helpful.
2. Be aware of the $\sim 1\ \frac{\mathrm{nH}}{\mathrm{mm}}$ series inductance. Keep supply and analog bonds as short as possible.
3. Provide sufficient spacing between PCB pads and ASIC pads.
4. Make the PCB pad large enough while meeting DRC rules.



Wire bonding: example I (Matt Cherry)



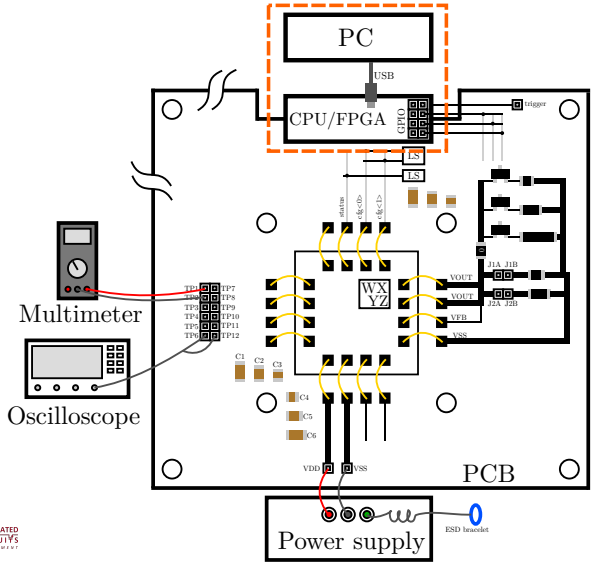
Wire bonding: example II (Matt Cherry)



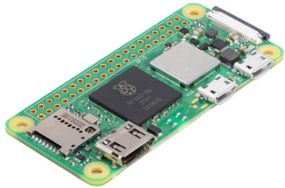
5

Embedded system

Embedded system



Raspberry Pi



Raspberry Pi Zero 2 W

★★★★★ (1 review) Write a Review

$17.25

Brand	Raspberry Pi
SKU:	1799
Condition:	New
Maximum Purchase:	1 unit
Model	Pi Zero 2

Essential extras:

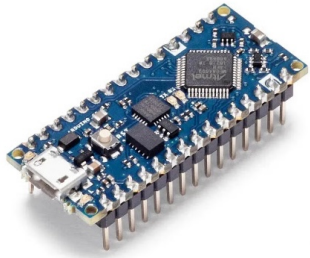
UniPiCase - Zero Standard

- A Linux system with GPIOs for 20$.
- Can run Claude on it!
- Do not use for deterministic / precise timing operations.
- Be careful about noise.

Raspberry Pi interface

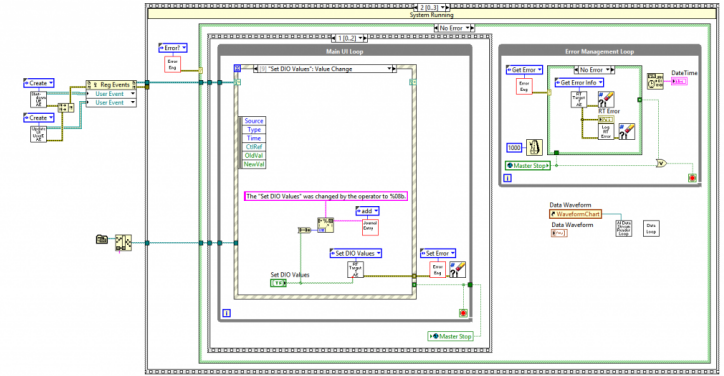
```
picocom -b 115200 /dev/cu.usbmodem11103
```

Arduino



- A microcontroller.
- C code gets mapped into assembly, so real-time operation.
- Not interactive by default.

LabVIEW-based DAQ

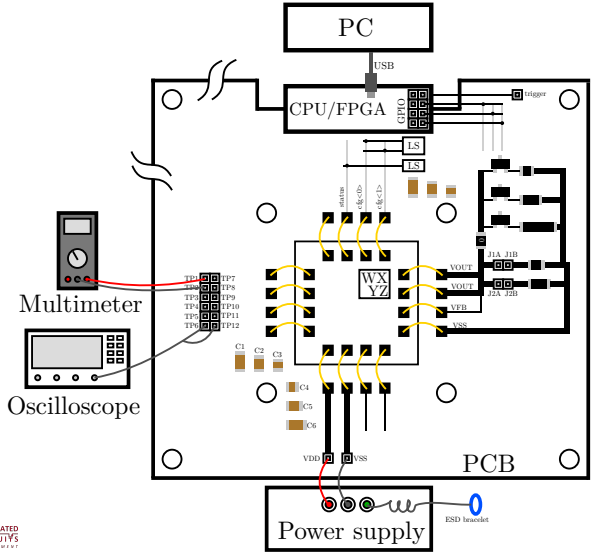


- Expensive.
- Could digitize the LDO output.
- Can synchronize events.

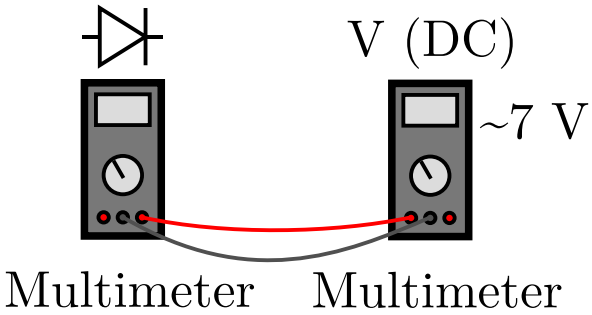
6

Sanity checks

Sanity checks

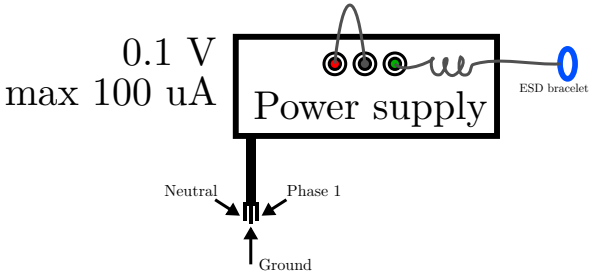


Sanity checks: short detection via multimeter



This can stress and damage the ASIC pads!

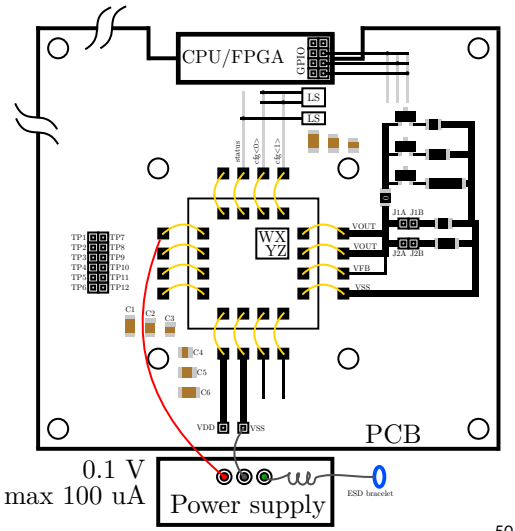
Sanity checks: short detection via power supply



To ensure the current protection works, short the supply and monitor the current.

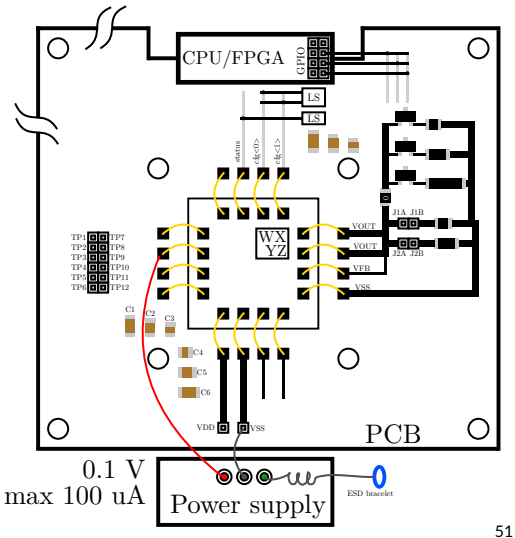
Sanity checks: short detection via power supply

- Monitor the current on the power supply.
- This also indicates if any VSS is open.
- Connections are actually done to the test points!



Sanity checks: short detection via power supply

- Monitor the current on the power supply.
- This also indicates if any VSS is open.
- Connections are actually done to the test points!
- Cycle through all the pads.

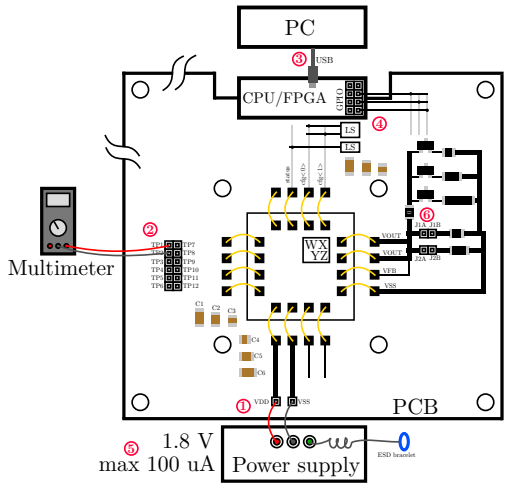


7

Power-up

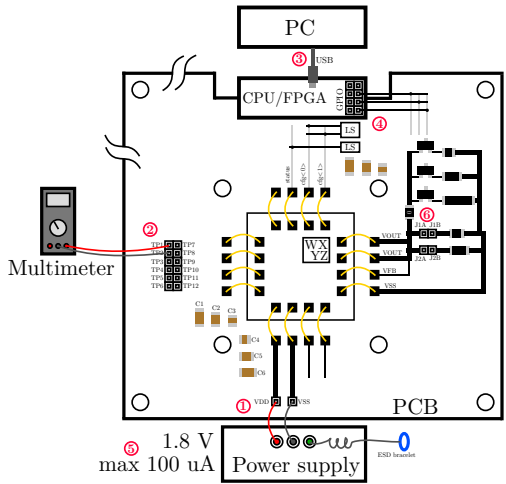
Power-up I

1. Connect the power supply with a conservative maximum current and nominal voltage.
2. Measure all the test points and internal nets.
3. Power your CPU.
4. Perform read and write operations to your embedded system.



Power-up II

5. Raise the current limit.
6. Exercise different loads and ensure a proper behavior.

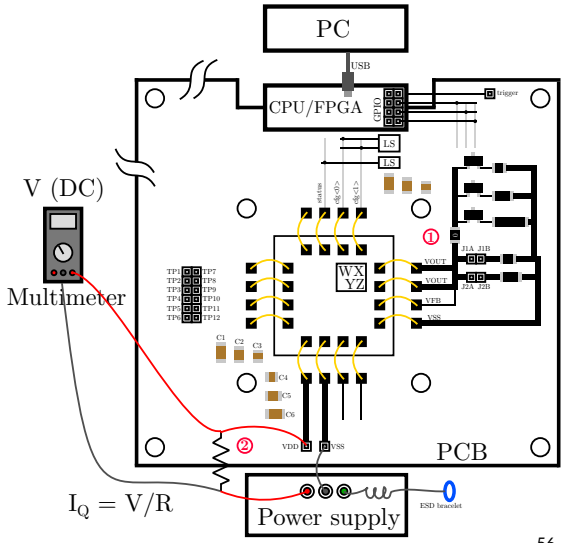


8

Characterization

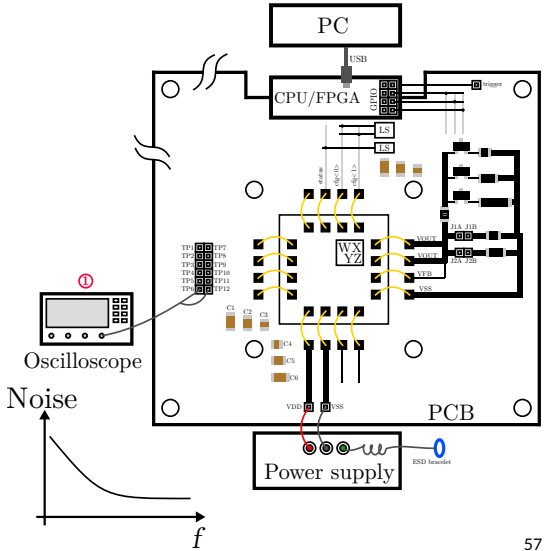
Characterization: I_Q

1. Ensure the output is floating (no load).
2. Pick a suitable resistor to get a measurable voltage drop, but not too large to power off the LDO.



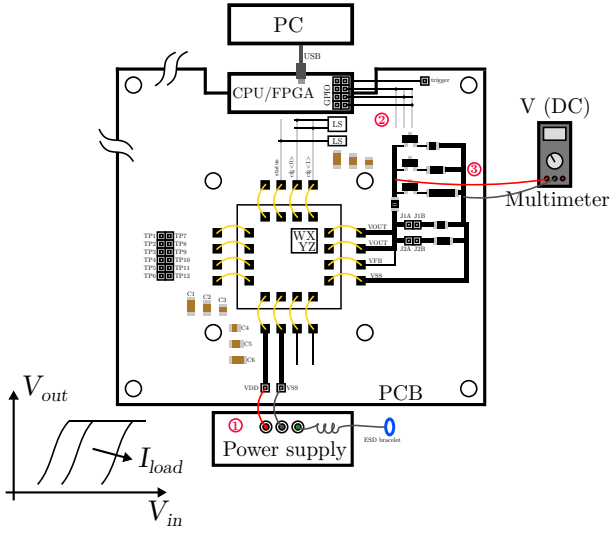
Characterization: noise

1. Perform an FFT on the transient waveform, report the voltage spectral density.



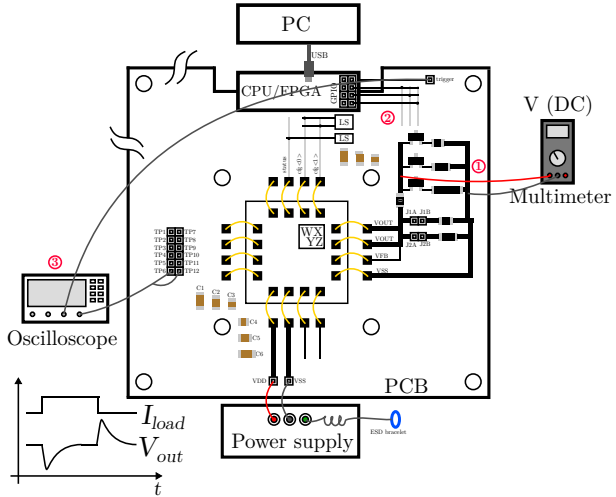
Characterization: minimum dropout [3]

1. Sweep V_{in}
2. Sweep I_{load}.
3. Measure V_{out}.



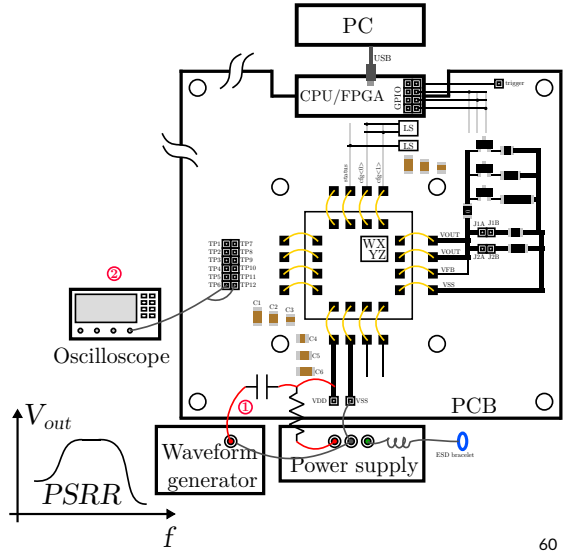
Characterization: load regulation

1. Measure in advance the loads precisely.
2. Toggle the load.
3. Capture the transient response. Can use the trigger signal.



Characterization: line regulation (PSRR)

1. Add an AC component on V_{in}
2. Measure the spectrum on V_{out}.



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[1] Karsten Kuhlmann. “Hochintegrierte aktive Sendeantenne für den Millimeterwellenbereich”. German. Advisor: A. F. Jacob. English title: “Highly integrated active antenna in the millimeter wave range”. Dissertation (Dr.-Ing.) Hamburg, Germany: Technische Universität Hamburg-Harburg, 2013. ISBN: 978-3-86387-264-9. DOI: 10.15480/882.1086. URL: https://hdl.handle.net/11420/1088.

[2] Hans Rosenberg. *The most misunderstood concept in decoupling*. YouTube. Accessed: 2026-08-31. URL: https://www.youtube.com/watch?v=TpXvac1Y3h0.

[3] Hyun-Sik Kim. “Fundamentals of Energy-Efficient LDO Regulator Design: From dropout optimization to wide-bandwidth and high-PSR techniques”. In: *IEEE Solid-State Circuits Magazine* 18.3 (2026), pp. 55–72. DOI: 10.1109/MSSC.2026.3696588.